

ONLINE SCHEDULING SYSTEM FOR ACADEMIC CONSULTATIONS USING DIJKSTRA'S ALGORITHM

Authors: CRUZ, MARK A. | DIZON, SHEILA R.

Year: 2024

Department of Information Technology

Degree Program: BSIT

Keywords: scheduling, Dijkstra, consultation system, graph algorithm, web system

ABSTRACT

This study presents an online consultation scheduling system that uses Dijkstra's algorithm to recommend the optimal available time slot for student-faculty consultations, reducing average wait time by 41%.

INTRODUCTION

Inefficient consultation scheduling at CvSU leads to faculty overloading on certain days and student frustration. Walk-in systems create bottlenecks, and informal messaging lacks transparency. This study develops a web-based system with an intelligent scheduling engine that models faculty availability as a weighted graph and recommends optimal consultation slots using Dijkstra's shortest path algorithm.

METHODOLOGY

The system was developed using Laravel (PHP) backend and Vue.js frontend, deployed on a local server. Faculty schedules are modeled as a directed weighted graph where edge weights represent time-slot priority scores based on faculty preference, room availability, and load balance. Dijkstra's algorithm computes the minimum-cost available slot for each consultation request. The system was pilot-tested with 15 faculty members and 120 students over one semester.

RESULTS

Average consultation wait time was reduced from 2.3 days to 1.35 days (41.3% reduction). Faculty reported 82% satisfaction with schedule fairness (Likert scale survey). System response time for slot computation averaged 0.4 seconds. No scheduling conflicts were recorded during the pilot period. Student adoption rate reached 78% by the 4th week.

DISCUSSION

Graph-based scheduling with Dijkstra's algorithm effectively optimizes consultation slot allocation. The system scales linearly with faculty count. Manual override capability was the most requested feature and was added in iteration 3. Future work should incorporate machine learning to predict preferred time slots based on historical scheduling patterns.